Networks of Queues and Jackson's Theorem

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Abstract

Tandem $M/M/1$, time reversibility and Burke's theorem, Kleinrock's independence assumption, Jackson's theorem, and ns-2 as the lab counterpart.

1 $M/M/1$ Queues in Tandem

Consider a simple system involving two transmission lines in tandem. The packet length doesn't change during transmission and transmission times are proportional to packet length. The packet lengths are exponentially distributed and arrivals occur at the first queue as a Poisson process. The first queue may be modelled as $M/M/1$, but the second queue cannot be. A packet which is large will take a long time in the first queue. During this time the second queue will have a chance to clear. The result will be that for the second queue larger packets are likely to arrive when the queue is smaller. This means that a large packet will typically experience less delay at the second queue than a small one. One analogy for this situation is a slow moving vehicle on a road. From this vehicle's driver's perspective there is very little traffic on the road because the slow moving vehicle never catches up with any. A similar thing occurs with $M/M/1$ queues in tandem. A large packet arrives at the second queue to find it has less packets because the smaller packets have had time to be processed. The effect of the first queue is to destroy the independence of the packet length and the state of the second queue. This problem is difficult to deal with analytically. To proceed with our analysis we will need to break this dependency. This can be achieved by considering service times for each packet to be statistically independent at each queue. We will now look at the consequences of this assumption for our simple pairing of two queues, but first we must introduce the concept of time reversibility.

2 Time Reversibility — Burke's Theorem

In the analysis of the Markov chain systems we have seen so far ($M/M/1$, $M/M/1/m$, $M/M/\infty$, $M/M/m$ and $M/M/m/m$) we relied on detailed balance to find the steady state distribution. Detailed balance equations

are a special case of global balance equations which arise when transitions only occur between neighbouring states in the Markov chain. Detailed balance can be applied to find the stationary distribution in any birth-death processes. In this section we will briefly mention an additional special property of birth-death processes (and other Markov systems for which detailed balance applies) known as *time reversibility.* The principle with time reversibility is as follows: for a birth-death process at steady state the arrival process is Poisson with rate λ. Time reversibility states that the departure process will also be Poisson with rate λ. We will now give some motivation for this statement, a proof is Given in Section 3.7 of [**?**]. Consider the following 'intuition' for the $M/M/1$ queue. The service rate for the system is given by μ. However this is not the departure rate because a departure can only occur when the system has a state $k > 0$. This condition is satisfied for a fraction of the time which is given by the utilization factor ρ. the departure rate is therefore given by $\mu\rho = \mu\frac{\lambda}{\mu} = \lambda$.

3 Burke's Theorem

Burke's theorem follows on from time reversibility. It states that if an $M/M/1$, an $M/M/m$ or an $M/M/\infty$ queue at its stationary distribution has a Poisson arrival process with a rate λ then the departure process is also Poisson with rate λ. Furthermore the number of the customers in the system at time t is *independent*of the sequence of the departure times prior to t. [1] The second property is worthy of comment as it sounds counterintuitive. You might expect that if there were a rapid series of departures from the queue then the queue would be shorter than normal, but Burke's theorem tells us that this is not the case. From this we can infer that the state of a queue *before*a rapid series of departures is about to occur must be abnormally high. Otherwise it *would* be abnormally low after the departures, but Burke's theorem tells us that this is not the case.

4 Kleinrock's Independence Approximation

Kleinrock's independence approximation suggests assuming independence for every node in the network. The approximation is good if large numbers of packets from outside the network are introduced at every queue. Consider a general network with labelled nodes and links between these nodes. There are several packet streams in the network, each stream has a rate of x_s. We define $f_{ij}(s)$ as the fraction of packets from stream s that cross the link between i and j. The rate associated with a link between node $i$and

$$jisthen$$

λ

[1] Queues with loss are slightly more complex because the arrival process does not have rate λ due to blocked packets.

4.1

*{Answer} From the diagram we can see which links traffic flows down. Let us take each link in turn. Using the superposition of Poisson flows we have

λ	1	2	3	4	5
1	-	-	-	-	-
2	$x_1 + x_2 = 15$	-	-	-	-
3	$x_3 = 7$	-	-	-	-
4	$x_4 + x_5 = 16$	-	-	-	-
5	-	$x_2 = 10$	$x_3 + x_6 = 15$	$x_4 + x_7 = 16$	-

$\rho = \lambda\bar{X}$	1	2	3	4	5
1	-	-	-	-	-
2	$15 \times 0.05 = \frac{3}{4}$	-	-	-	-
3	$7 \times 0.05 = \frac{7}{20}$	-	-	-	-
4	$16 \times 0.0333 = \frac{8}{15}$	-	-	-	-
5	-	$10 \times 0.05 = \frac{1}{2}$	$15 \times 0.05 = \frac{3}{4}$	$16 \times 0.033 = \frac{8}{15}$	-

$1 - \rho$	1	2	3	4	5
1	-	-	-	-	-
2	$\frac{1}{4}$	-	-	-	-
3	$\frac{13}{20}$	-	-	-	-
4	$\frac{7}{15}$	-	-	-	-
5	-	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{7}{15}$	-

$N = \frac{\rho}{1-\rho}$	1	2	3	4	5
1	-	-	-	-	-
2	3	-	-	-	-
3	$\frac{7}{13}$	-	-	-	-
4	$\frac{8}{7}$	-	-	-	-
5	-	1	3	$\frac{8}{7}$	-

The most congested links are from 1 to 2 and 3 to 5. To compute the answer to the second part we need the average delay at each link

$\bar{T} = \frac{\bar{X}}{1-\rho}$	1	2	3	4	5
1	-	-	-	-	-
2	$4 \times 0.05 = \frac{1}{5}$	-	-	-	-
3	$\frac{20}{13} \times 0.05 = \frac{1}{13}$	-	-	-	-
4	$\frac{15}{7} \times 0.033 = \frac{1}{14}$	-	-	-	-
5	-	$2 \times 0.05 = \frac{1}{10}$	$4 \times 0.05 = \frac{1}{5}$	$\frac{15}{7} \times 0.033 = \frac{1}{14}$	-

This means that to go via node 2 takes $\frac{1}{5} + \frac{1}{10} = 0.3$ via node 3 takes $\frac{1}{13} + \frac{1}{5} = 0.277$ via node 4 it takes $\frac{1}{14} + \frac{1}{14} = \frac{1}{7} = 0.1429$ according to Kleinrock's independence assumption.

4.2

*{Answer} If the service time distributions at the two queues are governed by exponential distributions with rate parameters μ_1 and μ_2 leading to utilization factors for each queue as $\rho_1 = \frac{\lambda}{\mu_1}$ and $\rho_2 = \frac{\lambda}{\mu_2}$. then the distributions for the states of the systems are given by

$$P(k_1) = \rho_1^{k_1}(1 - \rho_1)$$

$$P(k_2) = \rho_2^{k_2}(1-\rho_2).$$

The joint distribution for both systems is then

$$\begin{aligned} P(k_1, k_2) = & \quad P(k_1)P(k_2) \\ = & \quad \rho_1^{k_1}(1-\rho_1)\rho_2^{k_2}(1-\rho_2) \end{aligned}$$

because the two queues can be considered independently. As a result the average delay in the system can be computed as the sum of the average delays in each queue,

$$\bar{T} = \bar{T}_1 + \bar{T}_2 = \frac{1}{\mu_1 - \lambda} + \frac{1}{\mu_2 - \lambda}.$$

4.3 Thanks!

For more information on these subjects and more you might want to check the following resources.

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- podcast: The Talking Machines
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

4.4 Further Reading

- Burke, Kleinrock independence, and Jackson's theorem of **?**

4.5 Exercises

- Lab 2: tandem and multipath occupancy versus Kleinrock of **?**

References

Dimitri P. Bertsekas and Robert G. Gallager. *Data Networks*. Prentice-Hall, Upper Saddle River, NJ, second edition, 1992. ISBN 0-132-00916-1.

Neil D. Lawrence. Simulation of networks of queues in ns-2. Network Performance Analysis Lab Sheet 2, University of Sheffield, 2006. URL https://mlatcl.github.io/npa/.